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1. Write a C program to print the following sequence using a loop:

2 4 6 8 10 12 14 16 18 20

```

C loops.c X C loops2.c C whileloop.c
folderofc.c > C loops.c
1  #include<stdio.h>
2  int main(){
3      for(int i = 2; i<=20; i=i+2){
4          printf("%d", i);
5      }
6  }

```

2468101214161820

PS C:\Users\I C\VS\folderofc.c>

2. Write a C program to print the following sequence using a loop:

81 64 49 36 25 16

```

folderofc.c > C loops2.c
1  #include<stdio.h>
2  int main(){
3      for(int i = 9; i>=4; i--){
4          int square = i*i;
5          printf("%d\t", square);
6      }
7  }

```

816449362516

PS C:\Users\I C\VS

3. Write a C program that repeatedly asks the user to enter a number. Continue taking numbers until the user enters 0. For every number entered, calculate its cube and display the result. When the user enters 0, stop the process.

```
loops.c loops2.c whileloop.c cubeloo.c X
folderofc.c > C cubeloo.c
1  #include <stdio.h>
2
3  int main()
4  {
5      int num, cube;
6      while (1){
7          printf("Enter a number (0 to stop): ");
8          scanf("%d", &num);
9
10         if (num == 0){
11             break;
12         }
13         cube = num*num*num;
14         printf("Cube = %d\n", cube);
15     }
16     return 0;
17 }
```

```
Enter a number (0 to stop): 3
Cube = 27
Enter a number (0 to stop): 5
Cube = 125
Enter a number (0 to stop): 0
```

4. Write a C program that asks the user to enter the marks of a student. After entering the marks, ask the user whether they want to enter marks for another student, where 1 means Yes and 0 means No. Continue taking marks as long as the user chooses 1. At the end, display the marks entered and the total number of students.

```
C loops.c    C loops2.c    C whileloop.c X    C cubelooop.c    C temploop.c
folderofc.c > C whileloop.c
1  #include<stdio.h>
2  int main(){
3      int i=1,marks,sum=0;
4      while(i!=0){
5          printf("Enter marks: ");
6          scanf("%d",&marks);
7          sum = sum+marks;
8          printf("Do you want to enter more marks?:(1=yes,0=No)");
9          scanf("%d",&i);
10     }
11     printf("Total marks are: %d",sum);
12 }
13
```

```
Enter marks: 50
Do you want to enter more marks?:(1=yes,0=No)1
Enter marks: 30
Do you want to enter more marks?:(1=yes,0=No)0
Total marks are: 80
```

5. A weather application records the temperature for 7 days. Write a C program that asks the user to enter the temperature for each day. After taking all 7 temperature values, calculate the total temperature and count how many temperatures are greater than 100. At the end, display the total temperature and the number of temperatures greater than 100.

```
C loops.c    C loops2.c    C whileloop.c    C cubeloop.c    C temp
folderofc.c > C temploop.c
1  #include <stdio.h>
2  int main() {
3  int temp[7];
4  int sum = 0, count = 0;
5  for (int i = 0; i < 7; i++) {
6      printf("Enter temperature: ");
7      scanf("%d", &temp[i]);
8
9      sum = sum + temp[i];
10
11     if (temp[i] > 100) {
12         count++;
13     }
14 }
15 printf("Total temperature is: %d\n", sum);
16 printf("Temperatures greater than 100 is %d", count);
17 return 0;
18 }
19
```

```
Enter temperature: 50
Enter temperature: 24
Enter temperature: 80
Enter temperature: 90
Enter temperature: 110
Enter temperature: 32
Enter temperature: 4
Total temperature is: 390
Temperatures greater than 100 is 1
```

6. A student wants to save money every month. Write a C program that repeatedly asks the student to enter the amount saved. Continue taking amounts as long as the entered amount is greater than 0. Add each amount to the total savings and count the number of deposits. Stop when the user enters 0 or a negative value. At the end, display the total savings and the number of deposits.

```
loops.c loops2.c whileloop.c cubeloo.c
folderofc.c > saving.c
1  #include<stdio.h>
2  int main(){
3      int savedmoney=1,sum = 0,count = 0;
4      while(savedmoney>0){
5          printf("Enter saved money: ");
6          scanf("%d",&savedmoney);
7          sum = sum + savedmoney;
8          count++;
9      }
10     printf("The sum is: %d\n",sum);
11     printf("The number of deposits are: %d",count);
12     return 0;
13 }
14
```

```
Enter saved money: 50
Enter saved money: 32
Enter saved money: 8
Enter saved money: 0
The sum is: 90
The number of deposits are: 4
```

7. A customer is ordering food from a restaurant. Write a C program that asks the customer to enter the name of a food item and its price, then adds the price to the total bill. After each order, ask the customer whether they want to order another item, where 1 means Yes and 0 means No. If the customer enters 1, take the name and price of another item; if the customer enters 0, stop the ordering process. At the end, display the total bill and the number of items ordered.

```
loops.c    loops2.c    whileloop.c    cubelooop.c    temploop.c
folderofc.c > C orderloop.c
1  #include <stdio.h>
2
3  int main() {
4  char name[50];
5  int sum=0,price,choice;
6  do {
7      printf("Enter the name of food: ");
8      scanf("%s", &name);
9      printf("Enter price: ");
10     scanf("%d", &price);
11     sum = sum + price;
12     printf("Do you want to order more? (1 = Yes, 0 = No): ");
13     scanf("%d", &choice);
14
15 } while (choice == 1);
16 printf("The total bill is: %d\n", sum);
17
18 return 0;
19 }
20
```

```
Enter the name of food: pasta
Enter price: 500
Do you want to order more? (1 = Yes, 0 = No): 1
Enter the name of food: mayo
Enter price: 25
Do you want to order more? (1 = Yes, 0 = No): 0
The total bill is: 525
PS C:\Users\I C\VS\folderofc.c>
```

8. A company has 6 employees. Write a C program that asks the user to enter the salary of each employee and stores all 6 salaries in an array. After taking all salaries, use a loop to display the salary of each employee and count how many employees have a salary greater than 50,000. At the end, display all employee salaries and the number of employees whose salary is greater than 50,000

```
s.c      C loops2.c      C whileloop.c      C cubeloo.c      C templo.c      C sav
folderofc.c > C salaryloop.c
1  #include <stdio.h>
2
3  int main() {
4  int employees[6];
5  int count = 0;
6
7  for (int i = 1; i <= 6; i++) {
8      printf("Enter salary of employee %d: ", i );
9      scanf("%d", &employees[i]);
10
11     if (employees[i] > 50000) {
12         count++;
13     }
14 }
15 printf("\nEmployee Salaries:\n");
16 for (int i = 1; i <=6; i++) {
17     printf("Salary of employee %d is %d\n", i , employees[i]);
18 }
19 printf("Employees with salary greater than 50000: %d", count);
20 return 0;
21 }
22
```

```
Enter salary of employee 1: 50000
Enter salary of employee 2: 8000
Enter salary of employee 3: 98000
Enter salary of employee 4: 8900
Enter salary of employee 5: 78900
Enter salary of employee 6: 6789
```

```
Employee Salaries:
```

```
Salary of employee 1 is 50000
```

```
Salary of employee 2 is 8000
```

```
Salary of employee 3 is 98000
```

```
Salary of employee 4 is 8900
```

```
Salary of employee 5 is 78900
```

```
Salary of employee 6 is 6
```

```
Employees with salary greater than 50000: 2
```

```
PS C:\Users\I C\VS\folderofc.c> █
```